

Maximus Jialong Niu

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TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, SQL, C, C++

Frameworks & Libraries: React, Next.js, Tailwind, Storybook, Node.js

Databases: PostgreSQL (pgvector), DynamoDB

Developer Tools: Linux, Git, AWS (S3, IAM, Lambda), Azure (DevOps, Web Apps, AD B2C), Docker, Supabase, Jest

EMPLOYMENT

React Developer

Mar. 2025 – Present

Venuiti Healthcare

Waterloo ON, Canada

- Architected the frontend from scratch for the myMedcan healthcare portal, using **Next.js** and **TypeScript**, integrating a **Salesforce**-backed **REST API** across **88** endpoints to serve **20,000+** registered users
- Engineered a multi-step appointment booking and payment flow using **XState** and **React Query** to ensure predictable behaviour, integrating **Moneris** for billing, driving **15,000** monthly bookings and **\$168,000** in checkout transactions
- Integrated **Azure AD B2C** authentication on the frontend, eliminating client-side PHI exposure and ensuring **HIPAA** and **PIPEDA** compliance for **20,000+** users
- Built a library of **199+** reusable **React** components and hooks with **Storybook**, shared across multiple developers and pages and meeting client-mandated **WCAG AA** accessibility standards
- Configured an **Azure DevOps CI/CD** pipeline across development, staging, and production environments, automating containerized **Docker** builds and deployments to **Azure Web Apps** with an integrated **Jest** testing suite of **40+** unit tests that run on each deploy

Software Developer

Jan. 2023 – Apr. 2023

Wind River Systems

Ottawa ON, Canada

- Wrote and executed **50+** **C/C++** unit tests on a simulated **x86** board to validate the correctness of **33** Unix utility functions ported to the **VxWorks** RTOS
- Built an internal **Python** and **Flask** automation tool to convert between Markdown formats, processing **500+** pages and streamlining documentation workflows across teams

Full Stack Developer

Sep. 2020 – Dec. 2020

Syntu Software

Waterloo ON, Canada

- Architected a serverless system from scratch for handling sensitive legal evidence using **AWS DynamoDB**, **S3**, **Lambda**, and **IAM**, designed for **PIPEDA** compliance

Software Developer

Jan. 2020 – May. 2020

The Co-operators

Guelph ON, Canada

- Built an automation workflow utilizing **OCR** and **web scraping** to test an internal policy filing application, achieving **400+** policy submissions per hour
- Configured and executed **200+** automated regression tests for the policy filing tool, programmatically validating expected behaviour

PROJECTS

ReWatched – Semantic search over your entire YouTube watch history

- Built an **MV3-compliant Chrome extension** that silently indexes every YouTube video watched by fetching and chunking transcripts, embedding each chunk via **OpenAI**, and storing all vectors locally in **IndexedDB**
- Implemented fully client-side semantic search using **cosine similarity** against the stored index, returning ranked results with match confidence, thumbnails, and deep links that jump to the exact moment in the video

EDUCATION

University of Waterloo

Apr. 2024

Bachelor of Applied Science, Honours Computer Engineering

Relevant Courses: Software Architecture, Database Systems, Machine Learning, Systems Programming and Concurrency, Computer Networks, Computer Security, Compilers, Algorithms & Data Structures